

The background of the slide features a close-up, artistic rendering of numerous red blood cells. These cells are depicted as biconcave discs in various shades of red and orange, set against a dark, almost black, background. The cells are out of focus, creating a bokeh effect that emphasizes their circular shape and texture.

Management of patient with Arteriosclerosis and Atherosclerosis



Outlines:

Introduction

Pathophysiology

Risk factors

Clinical
manifestation

Complications

Diagnosis

Medical
management

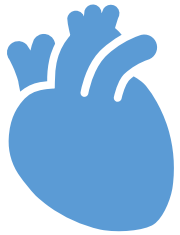
Surgical
management

Nursing
management

Teaching and Learning Method

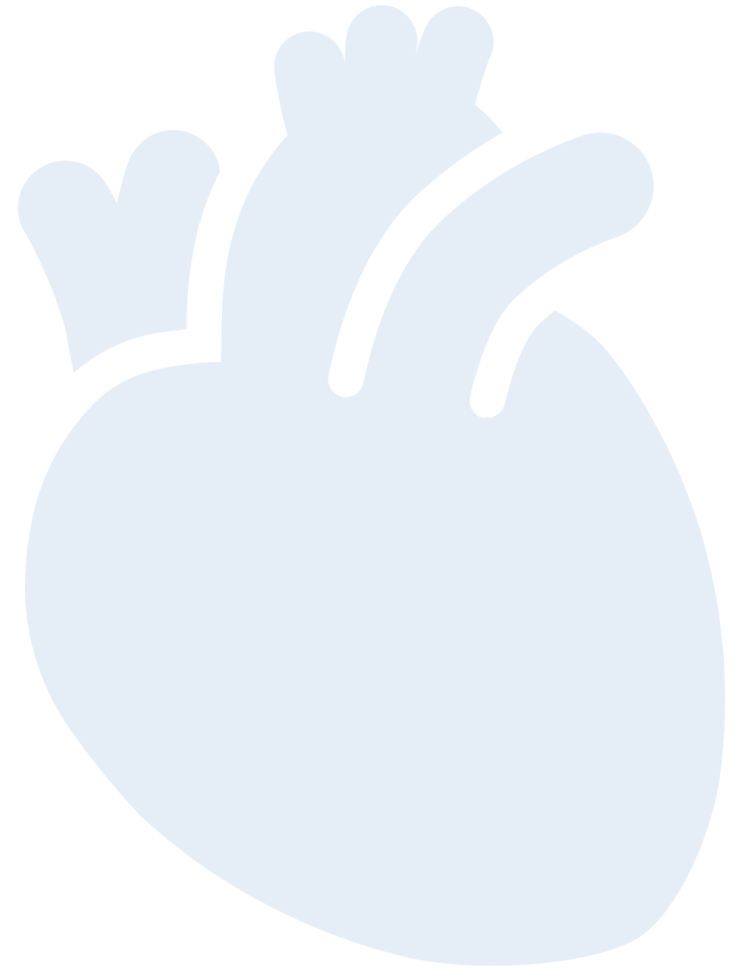
- Lecture
- Discussion
- Written assignments
- Case study



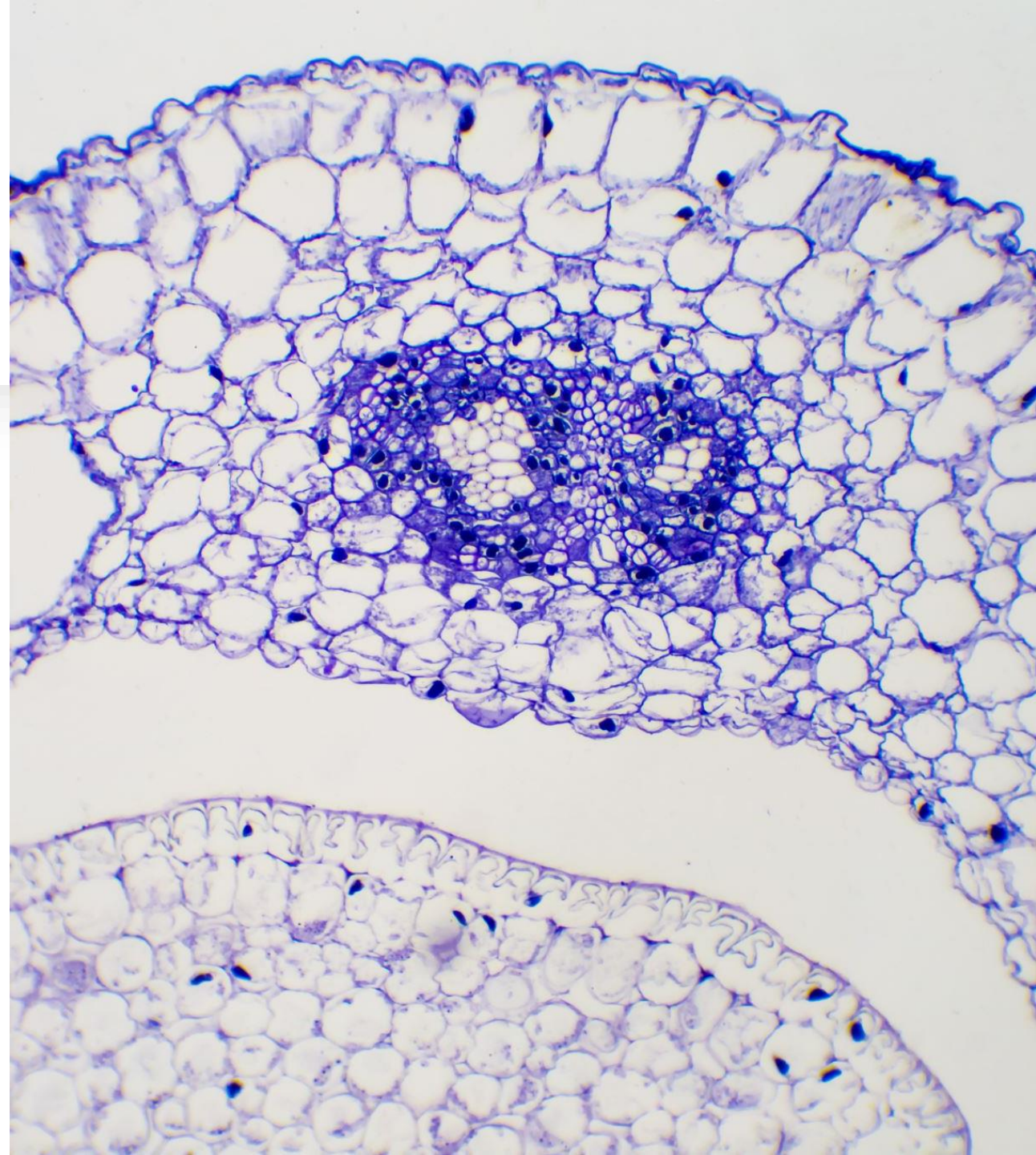


Arteriosclerosis

(hardening of the arteries) is a general term for a variety of arterial changes.

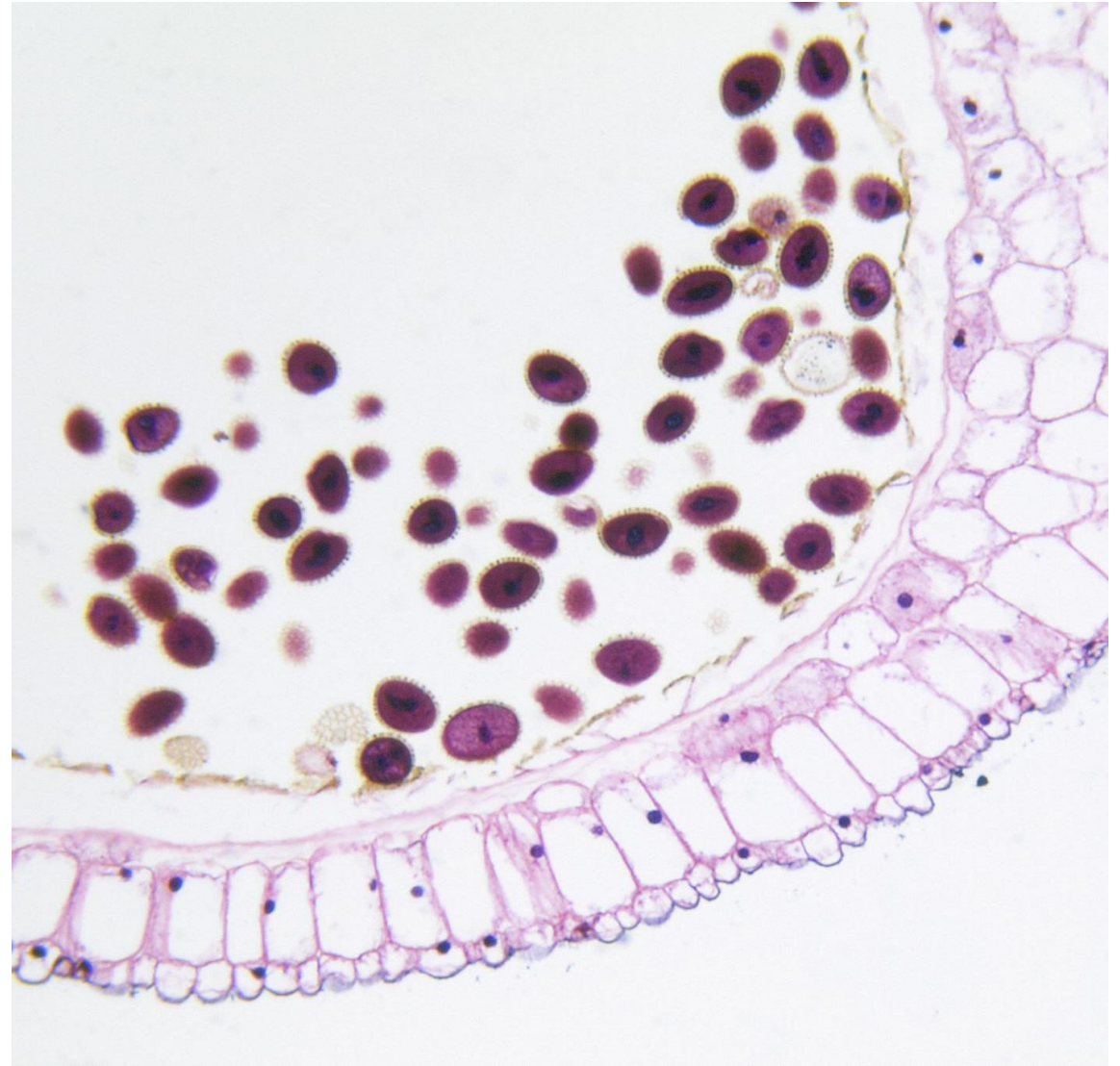


- ❖ Occurs with aging as degenerative changes occur in the small arteries and arterioles.
- ❖ Characterized by thickening of the artery walls that progresses to hardening as calcium deposits form. Vessel elasticity is lost.
- ❖ The thickening and calcification reduce the diameter of the vessels and cause slowing of blood flow, lead to ischemia and necrosis in various tissues.



Atherosclerosis form of artery narrowing.

- ✓ Lipids are deposited within the vessel walls and combine with cells, fibrin, and cell debris to form plaques.
- ✓ The plaque grows and extends into the lumen of the artery, where inflammation or erosion causes the plaque to be exposed to the blood.
- ✓ Uncontrolled Diabetes mellitus & Hypertension is a major factor in arteriosclerosis



Definition:

- Atherosclerosis is a chronic condition characterized by the buildup of fatty deposits (plaques) inside the walls of arteries, leading to thickening, hardening, and loss of elasticity of the arterial walls. This process narrows the arteries, restricting blood flow and potentially causing cardiovascular diseases such as coronary artery disease, stroke, and peripheral artery disease.

Causes:

High cholesterol levels: Elevated low-density lipoprotein (LDL) cholesterol contributes to plaque formation.



Hypertension (high blood pressure): Damages arterial walls, making them more susceptible to plaque buildup.



Smoking: Contributes to inflammation and damages blood vessels.



Diabetes mellitus: High blood sugar can damage arteries.



Obesity: Increases risk factors such as hypertension and diabetes.



Sedentary lifestyle: Lack of exercise leads to poor cardiovascular health.



Unhealthy diet: High intake of saturated fats, trans fats, and cholesterol.



Age: Risk increases with age.



Genetics: Family history of heart disease or atherosclerosis.



Chronic inflammation: From conditions like lupus or infections.

Signs and Symptoms:

- Atherosclerosis often progresses silently until significant artery narrowing or blockage occurs.

No symptoms in early stages.

- **Symptoms depend on affected arteries:**
 - **Coronary arteries:** Chest pain (angina), shortness of breath, fatigue.
 - **Carotid arteries (neck):** Sudden weakness, numbness, difficulty speaking, vision problems (stroke symptoms).
 - **Peripheral arteries (legs):** Leg pain while walking (claudication), numbness, coldness in extremities.
 - **Severe cases:** Heart attack or stroke.

Prevention:

- **Healthy diet:** Low in saturated fats, trans fats, cholesterol, and sodium; rich in fruits, vegetables, whole grains.
- **Regular exercise:** At least 150 minutes of moderate aerobic activity per week.
- **Quit smoking:** Avoid tobacco in any form.
- **Maintain healthy weight:** Prevent obesity.
- **Control blood pressure and diabetes:** Regular monitoring and medication adherence.
- **Limit alcohol consumption:** Moderate intake.
- **Stress management:** Techniques like meditation, yoga.

Treatment:

- **Lifestyle modifications:** Diet, exercise, smoking cessation.

Medications:

- Statins (lower cholesterol)
- Antihypertensives (control blood pressure)
- Antiplatelet agents (aspirin) to prevent clot formation
- Diabetes management drugs



Surgical interventions: For severe cases

- Angioplasty with stent placement
- Coronary artery bypass grafting (CABG)

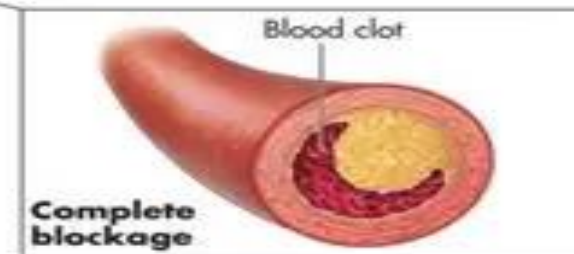
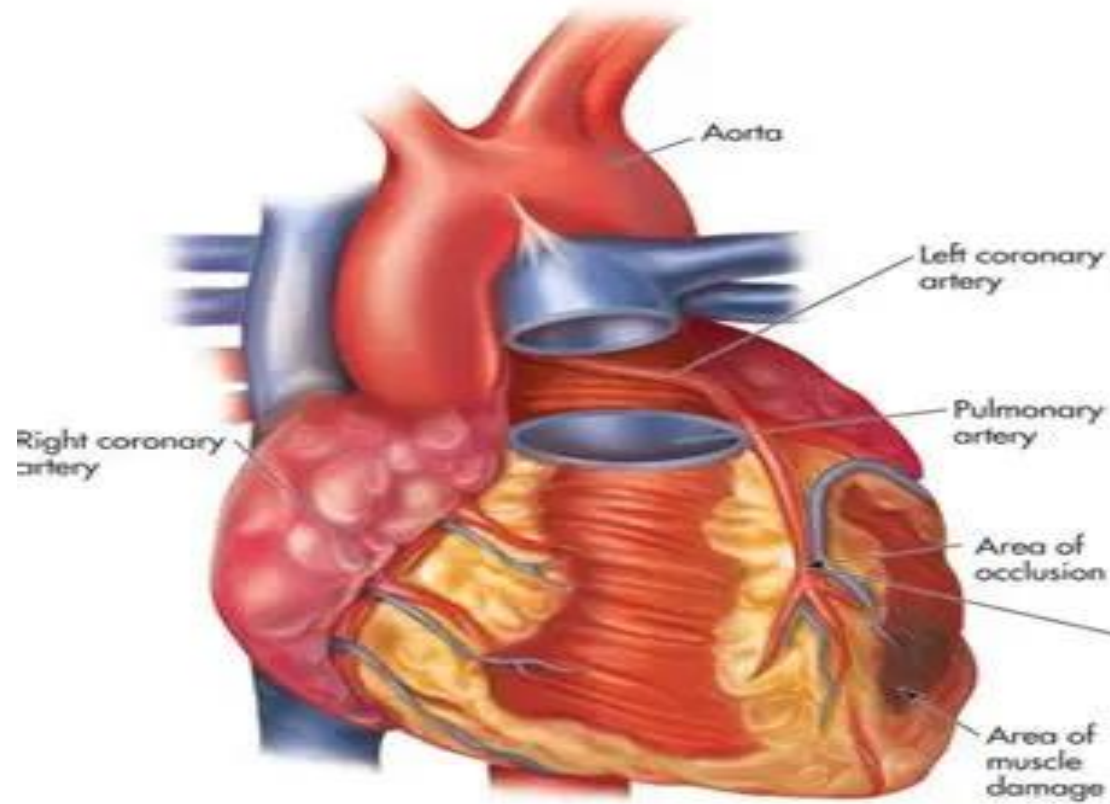
Regular monitoring: Blood lipid profiles, blood pressure, and cardiovascular health.



Nursing Care Diagnosis (Examples):

- **Ineffective Tissue Perfusion** related to arterial occlusion due to atherosclerosis.
 - **Risk for Decreased Cardiac Output** related to coronary artery blockage.
 - **Activity Intolerance** related to decreased oxygen supply to tissues.
 - **Imbalanced Nutrition: More than Body Requirements** related to unhealthy eating habits.
 - **Risk for Ineffective Health Management** related to lack of knowledge about disease and treatment.
 - **Anxiety** related to chronic illness and potential complications
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Atherosclerosis



Atherosclerosis

Progression



